Problem Statement #38 — Improved Source of Drinking Water

JalDrishti-AI: AI-Powered Drinking Water Access Analytics

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Domain: Water, Health & Sanitation / Public Policy / SDG Analytics
Dataset: AI Kosh — MIS 78th Round (nss78_21.xlsx) | **Survey Year:** 2023
Dataset Link: https://aikosh.indiaai.gov.in/web/datasets/details/improved_source_of_drinking_water_multiple_indicator_survey_78th_round.html
GitHub: https://github.com/varshiniparipelli/JalDrishti-AI

1. Problem Background

Access to safe and improved sources of drinking water remains a critical issue in India, especially in rural and underdeveloped regions. Despite ongoing efforts under the Sustainable Development Goals (SDGs), inequalities persist in water accessibility across states and socio-economic groups.

The MIS 78th Round data reveals key disparities: while the national improved water access stands at **95.7%**, states like **Kerala (55.2%)**, **Lakshadweep (73.8%)**, and **Manipur (75.6%)** lag severely behind. Piped water coverage is only **24.8% in rural areas** versus **61.5% in urban areas** — a critical equity gap requiring targeted policy intervention.

2. Objective

This project analyses data from the **78th Round of the Multiple Indicator Survey (MIS)** (AI Kosh dataset: nss78_21.xlsx) to:

- Assess the percentage of population with access to improved drinking water across 37 States & UTs
- Identify Rural–Urban and inter-state disparities in water access
- Predict water access levels using IBM Watson ML (GradientBoosting Regressor)
- Generate AI-powered policy insights using IBM watsonx.ai Granite model
- Track India's progress toward SDG 6.1 (universal safe water by 2030)
- Provide an evidence-based interactive policy dashboard for decision-makers

3. Dataset

Source: AI Kosh (India AI) — https://aikosh.indiaai.gov.in
Dataset Name: Improved Source of Drinking Water — Multiple Indicator Survey 78th Round
Direct Link: https://aikosh.indiaai.gov.in/web/datasets/details/improved_source_of_drinking_water_multiple_indicator_survey_78th_round.html
File: nss78_21.xlsx | **Survey Round:** MIS 78th Round | **Year:** 2023

Total Records: 222 rows | **States/UTs covered:** 37 | **Sectors:** Rural, Urban, All
Sub-indicators: (1) Improved Source of Drinking Water | (2) Piped Water into Dwelling or Yard/plot
Columns: state · age_group · sector · gender · indicator · sub_indicator · value · total_count
Value range: 0.7% – 100.0% | **Conducted by:** National Statistical Office (NSO), India

Column	Description	Example Values
state	State or Union Territory name	Bihar, Kerala, Delhi, All India
sector	Geographic sector	Rural, Urban, All
sub_indicator	Water access type measured	Improved Source · Piped Water
value	% of population with access	99.2 (Bihar Rural improved)
total_count	Survey sample reference	222

4. Key Findings (Real Data)

Indicator	Value
National Improved Water Access	**95.7%**
Rural Improved Water Access	95.0%
Urban Improved Water Access	97.2%
Piped Water — Rural	24.8%
Piped Water — Urban	61.5%
Rural–Urban Gap	2.2 percentage points
States above 90% access	31 out of 36
Lowest access — Kerala	55.2%
Highest access — Delhi, Chandigarh	100%
SDG 6.1 Projected 2030 coverage	100% — On Track
AI Model LOO MAE	3.98 percentage points

5. Solution — JalDrishti-AI

JalDrishti-AI is an end-to-end AI analytics platform built on **IBM Cloud Lite services** that analyses the real MIS 78th Round dataset and provides actionable policy insights through an interactive dashboard.

IBM Cloud Services Used

- **IBM watsonx.ai** — Model: ibm/granite-4-h-small · Project ID: d35a6f68-c129-4634-b823-ff9583f7fde6 · Generates national & state-level policy insights and powers the live AI chat
- **IBM Watson Machine Learning (Lite)** — GradientBoosting Regressor · LOO MAE: 3.98 pp · Predicts improved water access % per state & sector
- **IBM Cloud Object Storage (Lite)** — Stores nss78_21.xlsx in jaldrishti-data-bucket · ibm_boto3 SDK
- **IBM Db2 on Cloud (Lite)** — Table: MIS_DRINKING_WATER · SQL analytics · SSL port 50001
- **IBM Watson Assistant (Lite)** — Chatbot fallback · 10,000 messages/month · v2 session API

6. Technical Stack

- Backend: Python 3 + Flask (16 REST endpoints)
- Analysis: pandas, scikit-learn, joblib
- Frontend: HTML5 + CSS3 + Plotly.js (7 interactive charts)
- GitHub: https://github.com/varshiniparipelli/JalDrishti-AI

7. SDG Alignment

SDG 6.1: "By 2030, achieve universal and equitable access to safe and affordable drinking water for all." JalDrishti-AI directly supports this target by quantifying current coverage (95.7%), computing annual improvement rate (+3.21%/year), projecting 2030 coverage (100%), and identifying states requiring priority intervention.